

Probability of Tactical Nuclear Weapon Use in a US-Israeli Coalition Conflict with Iran

Assessment as of March 18, 2026

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This analysis constitutes a structured scenario-based exercise, not a prediction. Its value lies in identifying variables, triggers, and update indicators, rather than in specific percentages. The research directions outlined herein point toward a more methodologically rigorous analysis, which would, however, require institutional resources and an interdisciplinary team.

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1. Executive Summary

This analysis estimates the probability of tactical nuclear weapon (TNW) use by the US-Israel coalition in the ongoing conflict with Iran. The model employed is a working draft and uses a simplified Bayesian approach with odds ratio (OR) multipliers for identified input variables.

Key conclusion: even conservative model variants indicate that **the scenario of tactical nuclear weapon use cannot be automatically dismissed as entirely improbable**. However, the primary value of this analysis lies not in the numbers themselves, but in identifying escalation mechanisms, potential triggers, and points requiring further observation and in-depth analysis.

2. Operational Definitions

Definition of “tactical nuclear weapon use” (TNW): for the purposes of this analysis, ‘tactical nuclear weapon use’ refers exclusively to single or limited use, subordinated to an operational or signaling objective, rather than the initiation of a full-scale strategic exchange. The term encompasses the following variants, from least to most escalatory:

- a) **Atmospheric demonstration** (detonation over uninhabited terrain - ocean, desert - to signal readiness)
- b) **Strike on an isolated military target** (e.g., underground Fordow complex, missile launchers) using a low-yield warhead (<20 kt)
- c) **Warning shot** - detonation near a target without a direct hit, intended to produce a psychological and radiological effect
- d) **Strike on a military target in or near an urban area** (with anticipated civilian casualties - Scenario B in Section 9.2)

Each of the above variants constitutes a breach of the nuclear taboo maintained since 1945 and may trigger the cascade mechanisms described in Section 10.

Source reliability classification: in the conflict status tables (Sections 3.1-3.2) and throughout the document, the following classification system is used:

- a) **[Official]** - government announcements, UN agencies (IAEA), military commands (CENTCOM); highest formal reliability, but does not guarantee objectivity
- b) **[Wire/Agency]** - wire service dispatches (Reuters, AP, PAP (PL)) with verification methodology; high reliability, require ‘triangulation’ (confirmation of the same information by at least three independent sources)
- c) **[OSINT]** - open-source intelligence analysis (satellite imagery, social media, commercial data); moderate reliability, susceptible to disinformation
- d) **[Opinion]** - analyst assessments, expert estimates, commentary; low reliability without independent verification
- e) **[Model-based]** - parameters assigned by the author (OR, priors, P); reliability depends on the elicitation method (in this document: expert-based)

3. State of the Conflict

3.1. Status as of March 3, 2026

Calibration data from open sources [1][2][6][22][23][24]:

Parameter	Value	Source [Type]
Coalition strikes	~2,000 in 4 days	CNN [23], Al Jazeera [22] [Agency]
Decapitation strikes	Khamenei killed, 40+ commanders eliminated	CNN [23], PAP [6] [Agency]
Nuclear infrastructure	Natanz: damage to entrance buildings (IAEA: no radiological consequences expected)	IAEA via Reuters [1] [Official]
Iran's response	420+ missiles at 9 countries, Strait of Hormuz closure	PAP [6], Reuters [1][2] [Agency]
Regional escalation	Hezbollah enters the conflict (March 2)	Al Jazeera [22], Reuters [1][2] [Agency]
Trump's time window	4-5 weeks (public declaration)	Media [22][23][24] [Agency]
De-escalation signals	Iran - signals via diplomatic channels to Gulf states	OSINT / unconfirmed [OSINT]

3.2. Update - status as of March 18, 2026

Calibration data from open sources [1][2][3][4][5][6][7][22][23][24][33][43][44]:

Parameter	Value	Source [Type]
Casualties in Iran	At least 1,444 killed, 18,551 wounded (Iran Health Ministry)	PAP [6] [Wire] - data should be treated as "at least"
Decapitation strikes	Ali Larijani, Gholamreza Soleimani (Basij), Esmail Khatib (intel. min.) - killed	Reuters [1][2][7], AP [33] [Agency]
Iran's missile capabilities	Decline of >90% (350 missiles/day → ~25/day); 80% of strike capability against Israel destroyed	Coalition-side assessments [OSINT/Opinion] - no independent verification
U.S. air dominance	B-1 non-stealth over Iran; >5,000 targets struck (CENTCOM, March 10)	CENTCOM / DoD [4] [Official]
Casualties in Israel	At least 15 killed, >3,369 wounded; 2 casualties in Ramat Gan 03/18	Reuters [1][2], PAP [6] [Agency]
U.S. casualties	13 servicemembers killed in the region (Iranian attacks)	DoD [4] [Official]
Regional escalation	Iranian attacks on UAE, Qatar, Kuwait, Saudi Arabia, Bahrain; Israel bombs Beirut	Reuters [1][2], 18.03.2026 [Agency]
European allies	Germany and EU refuse participation; UK provides Diego Garcia and RAF Fairford bases	UK Gov [3] [Official]

Parameter	Value	Source [Type]
Diplomacy	Trump refuses negotiations; doubts whether Mojtaba Khamenei controls the situation	Reuters [1][2], media [22][23][24] [Agency /OSINT]
China	Trump delays Beijing visit; asks China for help reopening Strait of Hormuz	Reuters [44], FT [Agency]
Resignation in U.S. admin.	Joe Kent (intelligence) “resigns”: “Iran did not pose a direct threat”	AP [33] [Agency]
IAEA – Isfahan	IAEA does not know the status of the new underground enrichment facility in Isfahan without inspection (Grossi)	Reuters/IAEA [2] [Official]
U.S. public opinion	27-29% approval, 43% disapproval of strikes (Reuters/Ipsos, 02/27-03/01)	Reuters/Ipsos [5] [Wire - with methodology]
China’s defense budget	+7% y/y (2026) - highest ever	Reuters [43] [Agency]

MODEL IMPLICATION: The operation is proceeding “according to the coalition’s plan,” or at least according to official reports / assurances from its representatives: Iran’s missile capabilities have declined by >90%, air dominance has been achieved. However, the absence of diplomatic de-escalation and regional escalation (attacks on Gulf states) increase the risk of the conflict extending beyond “Trump’s window.”

NOTE: The claim of a “>90% decline in missile capabilities” originates from coalition/OSINT sources and has moderate/low reliability without independent verification. The IAEA describes Natanz in cautious language (“damage to entrance buildings”), while coalition documents use the stronger “significantly damaged.” This discrepancy illustrates the need for source-type tagging.

4. Analytical Framework

4.1. Nuclear Doctrines

USA (Nuclear Posture Review 2022, in force; NDS/NPR/MDR 2022) [8]: No no-first-use declaration. Tactical use possible in cases of: (1) WMD use by an adversary, (2) catastrophic failure of conventional operations, (3) existential threat to an ally. The Pentagon has not updated the NPR for 2026 (Air & Space Forces, March 2026) [9].

Israel: Two pillars (Narang 2014 [10], Cohen 2010 [11]): Samson Option (ultimate retaliation under existential threat) and Begin Doctrine (conventional preemptive strikes - currently being implemented, but without nuclear weapons).

4.2. Escalation Ladder (Schelling 1960/1966 [12][13], Bracken 1983/2012 [14][15])

Current level:

- [✓] Massive conventional strikes
- [✓] Missile exchange + decapitation strikes
- [✓] Multi-front warfare (Hezbollah, Yemen, Iraq)
- [–] Chem/bio weapons use
- [–] Nuclear device demonstration by Iran
- [–] Existential threat to Israel
- [–] **Tactical nuclear weapon ← several rungs above**

5. Model Variables

5.1. Trigger Scenarios (event tree model)

Trigger	Wk. 2	Wk. 4	Wk. 6	P(nucl. trig.)
Iran uses chem/bio weapons against U.S. or Israeli forces	4%	8%	12%	~10-15%
Iran demonstrates nuclear capability	1%	2%	3%	~25-30%
Existential threat to Israel	3%	6%	9%	~28-35%
Conventional operation failure + political pressure	2%	4%	7%	~3-5%
Defensive breakthrough: interceptors exhausted	3%	10%	18%	~20-30%

5.2. Structural Factors (OR multipliers): author's selection

Multiplicative model: $\text{odds_final} = \text{odds_baseline} \times \text{OR}_1 \times \text{OR}_2 \times \dots \times \text{OR}_n$

Factor	OR (Wk. 6)	Rationale
Historical precedent (Hiroshima/Nagasaki 1945)	×1.5	Only country with precedent
Interceptor depletion + low-cost strike means	×2.0	JINSA [25], Fortune [26], DSA [27]
Diversionary escalation mechanism ('wag the dog' hypothesis / "Epstein Effect")	×1.5	Wag the dog effect
Signaling toward China/Russia	×2.0	Hiroshima→USSR analogy
Trump administration (norms)	×1.5	"Nothing off the table"
Reputational risk (no progress = failure)	×1.4-1.8	Active wk. 4-6
Taiwan threat + time pressure (Davidson Window 2027)	×1.8-2.2	NEW VARIABLE [7][31][36]

METHODOLOGICAL NOTE ON THE OR TABLE: (1) "Epstein Effect" (wag the dog) - operational indicator: >30% decline in media attention to U.S. domestic affairs (e.g., Epstein, criminal proceedings, scandals) in the week following military escalation. (2) Variables V4 (China/Russia Signaling) and V7 (Taiwan) potentially overlap mechanistically - both relate to signaling value toward China. The multiplicative model may overestimate their combined impact. A more conservative approach: treat them as a single variable with OR = 2.5-3.0 instead of two separate ones (2.0 × 2.0). (3) ORs are intuitive and would require formal expert elicitation for greater reliability. All ORs are classified [Model-based].

5.3. New Variable: Taiwan Threat

Input data (March 2026):

- a) China defense budget 2026: USD 278 bn (+7% y/y), highest ever (Reuters, 03/05/2026 [43])
- b) Davidson Window: China expected to achieve full invasion capability by end of 2027 (USNI News, 2021 [36])
- c) China actively drills Taiwan blockade (Japan Times, March 2026 [31])
- d) US Intelligence Assessment: China is not currently planning to invade Taiwan by 2027 (Reuters, 03/18/2026 [7])
- e) China learns from the Iran war: AI, tactics, depletion of U.S. stockpiles (Global Times, AEI [29] [OSINT/Opinion])
- f) Pentagon Report 2025: China making progress despite personnel purges (DoD, December 2025 [30])

Influence mechanism: The U.S. faces a two-front dilemma. A protracted conflict with Iran depletes resources (interceptors, precision munitions) essential for Taiwan's defense. **At the same time, a dramatic conclusion to the conflict (nuclear strike) would serve as a deterrence signal to China: "If we do this to Iran, we'll do it to you."** A single action achieves two strategic objectives.

OR = 1.8-2.2 (high, as it combines time pressure with signalling value) [Model-based]

5.4. Disaggregation of the "interceptors" node into sub-nodes

The "interceptor depletion" node (OR = 2.0) in the baseline model functions as a "black box." For greater transparency and updatability, it should be disaggregated into three sub-nodes:

Sub-node	Description	Data availability	Implication for OR
(a) Stockpile status	Current interceptor stockpile levels (Iron Dome, Arrow, David's Sling, THAAD, Patriot)	Very low - no public data; high disinformation risk	If stockpile <30%: sub-node OR ↑
(b) Consumption rate	Daily consumption vs. Iranian attack intensity (~25/day post-degradation vs. 350/day initially)	Low - derived from coalition claims [OSINT]	Declining after possible Iranian degradation
(c) Replenishment rate and production constraints	RTX/Raytheon production, allied deliveries (Japan, Korea); lead time	Moderate - industry data JINSA [25], Fortune [26], DSA [27] [Wire/OSINT]	Key: if lead time >4 wk., OR rising in wk. 4-6

Recommendation: each model update should separately assess each sub-node. OR = 2.0 for the entire node should be treated as a weighted average of the three sub-nodes. Under current information conditions (no hard data on stockpile levels), this node carries the highest uncertainty in the entire model.

6. Results

6.1. Final Calculation (multiplicative model)

Baseline wk. 6 (without factors): ~3% → odds ≈ 0.031*

Converting probability to odds:

Multiplicative model operates on odds, not probabilities, because OR (odds ratio) multipliers by definition act on odds, not on P. The conversion between these quantities is as follows:

odds = $P / (1 - P)$, where P is the probability of the event, and (1 - P) is the probability of its non-occurrence. Odds express the ratio of "chances for" to "chances against." For example: P = 50% → odds = 0.5/0.5 = 1.0 (even odds); P = 75% → odds = 0.75/0.25 = 3.0 (threefold advantage "for").

For baseline $P_0 = 3\%$ (i.e., 0.03): **odds₀ = $P_0 / (1 - P_0) = 0.03 / (1 - 0.03) = 0.03 / 0.97 \approx 0.031$**

$0.031 \times (1.5 \times 2.0 \times 1.5 \times 2.0 \times 1.5 \times 1.6 \times 2.0) = 0.031 \times 43.2 \approx 1.34$

$P = 1.34 / (1 + 1.34) \approx 57\%$

The value 0.03 is the adopted probabilistic baseline (see Sec. 8.4); 0.97 is its complement to unity. The result 0.031 means that without any escalation factors, for every ~32 scenarios in which nuclear weapons are not used, there is ~1 scenario in which they are.

After multiplying odds₀ by the product of all ORs to obtain the final odds, the reverse conversion to probability: **$P = \text{odds} / (1 + \text{odds})$**

The method "prior odds × OR multipliers = posterior odds" is the standard form of Bayes' theorem expressed in odds form. The most authoritative and directly relevant sources:

1. Already in this document [39]: MIT OCW 18.05 (Orloff & Bloom). Quote from it: **"The prior odds of H are O(H). The posterior odds after D₁ are $O(H|D_1) = BF_1 \cdot O(H)$ "** (https://ocw.mit.edu/courses/18-05-introduction-to-probability-and-statistics-spring-2022/mit18_05_s22_class12-prep-b.pdf) - and further shows that under conditional independence of evidence, the posterior can be computed sequentially by multiplying successive Bayes factors. **The method used in this model is the odds form of Bayes' theorem** (not merely "OR algebra"): prior odds are updated by multiplying successive odds ratios, each functioning as a Bayes factor for the corresponding variable.

2. Heuer, R.J. (1999). *Psychology of Intelligence Analysis*. CIA/CSI. Heuer explicitly recommends in intelligence analysis: **"insert an odds ratio or probability range in parentheses after expressions of uncertainty in key judgments."** (https://www.bmbs.org/febrero/1D_Psychology_of_Analysis_EXCERPTS.pdf) This provides methodological context: Bayesian updating in odds form is standard practice in strategic and intelligence analysis.

3. Formal textbooks:

- a) **Downey, A.B. (2013). *Think Bayes*. O'Reilly.** Ch. 5 "Odds and Addends" - directly: **"the posterior odds are the prior odds times the likelihood ratio. This is the odds form of Bayes's theorem."** <https://www.greenteapress.com/thinkbayes/html/thinkbayes006.html>
- b) **Taboga, M. (2021). *Lectures on Probability Theory and Mathematical Statistics*. StatLect** - formal definition: **"the posterior odds ratio between M₁ and M₂ quantifies how much more likely M₁ is than M₂, based on the prior and the evidence provided by the data"** <https://www.statlect.com/fundamentals-of-statistics/posterior-odds-ratio>

Horizon	Conservative estimate	Full model estimate	Dominant scenario
2 weeks	6-8%	10-15%	Chem/bio + Samson Option
4 weeks	17-22%	30-40%	+ interceptors + reputation
6 weeks	27-32%	55-70%	+ Taiwan + interaction effects

NOTE: Even the lower bound (27-32%) at week 6 of hostilities represents a risk level comparable to the 1962 Cuban Missile Crisis. Upper estimates (55-70%) compel treating the event as a planning contingency, not a marginal scenario.

7. Model Update Conditions

7.1. Conditions for Lowering Probability

The model should be updated DOWNWARD if by week N the following occur:

Event	Adjustment	Condition
Iran formally signals capitulation / negotiations	-5-10 pp*	By wk. ~3
Hezbollah withdraws from the conflict	-3-5 pp	By wk. ~3
Interceptors replenished by allies (Japan, Korea)	-4-6 pp	By wk. ~4
China publicly warns U.S. against nuclear escalation	-2-4 pp	Anytime
U.S. Congress introduces new restrictive limitations on nuclear weapons use	-5-8 pp	Anytime
Operational objectives achieved before Trump's window expires	-10-15 pp	By wk. 4-5

*pp = percentage points (change expressed in absolute terms, not relative - e.g., a decline from 57% to 45% is -12 pp, not -12%)

7.2. Conditions for Raising Probability

The model should be updated UPWARD if by week N the following occur:

Event	Adjustment	Condition
Iran uses chemical/biological weapons	+15-25 pp	Anytime
Iran demonstrates nuclear capability	+20-30 pp	Anytime
Mass civilian casualties in Israel (>1,000) / critical infrastructure destruction	+10-20 pp	Anytime
Coalition interceptors fall below 20% of stockpile	+8-12 pp	By wk. 4
China announces exercises around Taiwan during the conflict / Chinese attack on Taiwan	+5-8 pp	Anytime

8. Methodological Notes

8.1. Model Limitations

- a) ORs assigned intuitively, not based on formal expert elicitation [Model-based]
- b) Interaction effects between variables are not fully quantified
- c) The probabilistic baseline (3%) requires further validation (see Sec. 8.4)
- d) The proper model is a Bayesian network with inter-node correlations (see Sec. 16.2)
- e) The nodes “China/Russia signaling,” “Taiwan,” “reputation,” and “time window” likely partially describe the same strategic-political pressure mechanism - a classic “double counting” risk scenario

8.2. Factors Not Included (potential inhibitors)

- a) Conventional response by Russia (e.g., intensification in Ukraine)
- b) Secondary sanctions / embargo
- c) Accelerated proliferation (North Korea, Saudi Arabia)
- d) Reputational cost of nuclear weapon use (international isolation)

8.3. Epistemic Caveats

Semi-precision: The numbers presented in this analysis (e.g., “27-32%,” “55-70%”) lack a solid empirical basis, and in the author’s view - regardless of who performs a similar analysis - will always be subjective. Odds ratios are intuitive, not calibrated. The probabilistic baseline (3%) is arbitrary. An honest version should read: the risk appears significantly higher than the historical norm (1-3%), likely in the range of serious to very serious, but we cannot provide a number with confidence.

Lack of real-time falsifiability: The model was not written before events, but during them. The “update conditions” section is post hoc. A true predictive model should specify in advance which observations would lower or raise estimates; however, this model identifies the peak probability of tactical nuclear weapon use by the coalition around week 6 from the start of hostilities, **thus still qualifying as a prediction.**

VERDICT: This analysis is a scenario exercise, but the numbers should not be treated as predictions. Nobody knows what the actual probability is - even the people in the Situation Room are guessing, just with better data.

8.4. Validation of the Probabilistic Baseline (prior 3%)

The prior $P_0 = 3\%$ at week 6 is the parameter with the highest impact on the model’s final output. Changing it by ± 1 pp shifts the final P by over a dozen percentage points. The source of this value requires explanation:

- a) **Source:** The 3% prior is an author’s estimate [Model-based], based on analogy with historical nuclear crises (Cuban Missile Crisis 1962, Able Archer 1983) and general reasoning about the possibility of nuclear weapon use in conventional conflicts.
- b) **Historical validation:** The International Crisis Behavior Project (ICB, Duke University / University of Southern California) database [21] contains >470 international crises since 1918. None of them ended with nuclear weapon use after 1945. A naive historical estimate would therefore be $0/\sim 470 \approx 0\%$. However, (a) most crises did not involve nuclear states, (b)

none involved direct massive strikes on the nuclear infrastructure of a threshold state.

Absence of precedent does not mean zero probability.

- c) **Prior sensitivity:** At $P_0 = 1\%$, the model yields final $P \approx 30\%$; at $P_0 = 5\%$, the model yields $P \approx 69\%$. Prior dominance is the most significant weakness of the multiplicative model.

CONCLUSION: The 3% prior is not formally validated. It should be treated as a “reasonable midpoint” between the baseline historical frequency (~0%) and an intuitive estimate for an unprecedented situation. Formal validation would require an expert panel (Cooke’s/SEJ method [19][20]) - see Sec. 16.2.

9. Analysis of TNW Use Consequences by the Coalition - conditional scenarios

METHODOLOGICAL NOTE: The sections below have been rewritten as conditional scenarios with mechanisms and ranges, in line with review recommendations. Each scenario specifies an entry condition, causal mechanism, range of consequences, and literature reference. The sections draw on the Sagan-Waltz proliferation debate [41] (Sagan 1996/2012; Waltz 1981/2012) and Tannenwald’s work [42] (1999/2007) on the nuclear taboo.

Neither scenario (A or B) accounts for the CASCADE EFFECT described in Section 10 (‘conditional scenario C’).

9.1. Scenario A: Uninhabited Target

Condition: Detonation over uninhabited terrain (atmospheric demonstration), strike on an underground military complex (e.g., Fordow), or isolated launchers.

Mechanism: The coalition leverages the “no civilian casualties” argument, presenting the use as a “surgical, ultimate solution.” Military objective achieved. Nuclear taboo weakened but not destroyed - a new category emerges: “acceptable tactical use” (cf. Tannenwald 2007 [42] on conditions for taboo erosion).

Short-term consequences (0-3 months):

- a) Military objective achieved; Iran loses capacity to continue nuclear program
- b) Coalition’s diplomatic isolation limited - some allies maintain neutrality, but others announce intent to leave the alliance with the U.S.
- c) USA: domestic polarization, protests; Trump frames it as a “surgical, absolutely necessary solution”
- d) Israel: deterrence doctrine strengthened; government on the verge of collapse

Medium-term consequences (3-5 months):

- a) NATO fractures functionally - Europe divided; AUKUS and Japan closer to the U.S.
- b) China accelerates armament; Davidson Window shortens
- c) **Proliferation (Sagan scenario):** 15-20 new states initiate nuclear programs within a decade (vs. current 9 nuclear states). Sagan (1996) [41] argues that more nuclear states = more failure points (organizational failures, civil-military tensions). Waltz (1981/2012) [41] counters that deterrence works regardless of the number of actors
- d) **In both the U.S. and Israel, governments “fall” as a result of protests**

9.2. Scenario B: Inhabited Target

Condition: Strike on Tehran (military/government districts with civilian casualties), Isfahan, Shiraz; or “accidental” mass casualties from a strike on a military target in an urban area.

Mechanism: Full breach of the nuclear taboo (Tannenwald 2007) [42]. Threshold states interpret this as proof that nuclear weapons are used - not merely possessed - which radically alters the proliferation calculus.

Short-term consequences (0-3 months):

- a) Total diplomatic isolation - even the UK and Australia publicly distance themselves
- b) ICC issues arrest warrants; the U.S. and Israel become pariah states *de iure*
- c) USA: mass protests; impeachment possible but unlikely with a Republican Congress
- d) Israel: international delegitimization; BDS movement goes mainstream

Medium-term consequences (3-5 months):

- a) NATO disintegrates functionally - Germany, France exit command structures
- b) China: strategic opportunity - presents itself as a “responsible power”; gains Global South
- c) **Nuclear taboo destroyed:** threshold for future conflicts virtually nonexistent (cf. Tannenwald 1999: taboo as a constitutive norm - its breach signifies a change in identity, not merely behavior)
- d) **Proliferation (extreme scenario):** 20-30 states initiate nuclear programs within 10 years. This scenario falls on Sagan’s side of the Sagan-Waltz debate: institutional weaknesses of new nuclear states (lack of safety procedures, civilian control, political instability) dramatically increase the risk of accidental or unauthorized use.
- e) Economy: global recession, de-dollarization accelerates, deglobalization

9.3. Regardless of Scenario: End of the Post-War status quo

The difference between Scenarios A and B is qualitative, not gradual. However, even Scenario A ends the post-Cold War order built on three pillars:

- a) **NPT (Non-Proliferation Treaty = Treaty on the Non-Proliferation of Nuclear Weapons, 1968) as a nonproliferation regime** - proof that non-nuclear states are defenseless against great powers
- b) **Nuclear taboo since 1945** - completely broken after 81 years [42]
- c) **USA as a guarantor of stability** - transformation into a revisionist actor employing weapons of mass destruction

CONCLUSION: Any use of tactical nuclear weapons by the coalition, regardless of the scale of casualties, opens an era of multipolar nuclear anarchy. The difference between scenarios concerns the pace and depth of order destruction, not the fact of destruction itself.

10. Cascade Effect: Global Precedent ('conditional scenario C')

Entry condition: **USA/Israel uses TNW against Iran (any variant from Section 2). The precedent acts as a mechanism triggering a sequence of decisions in other regions of the world.**

10.1. Russia and Ukraine

Mechanism: USA/Israel uses TNW → Russia gains immediate precedent and rhetorical legitimization → Putin: "The West broke the taboo itself. We are merely defending our territory" (Crimea, annexed oblasts) → **Russia strikes tactically in Ukraine within days/weeks.**

P(Russia uses in Ukraine | USA used on Iran) ≈ 30-50% within 30 days [Model-based]

UK (and/or France) response¹ - correction regarding the Budapest Memorandum: The UK is a signatory of the Budapest Memorandum (1994), which contains security assurances for Ukraine - **not an automatic obligation of armed defense** (unlike NATO treaty obligations, Art. 5). The Memorandum obliges signatories to (a) refrain from using force against Ukraine, (b) refrain from economic threats, (c) consult in the event of a threat. It does not contain a mutual defense clause (cf. Budjeryn 2022; CISAC Stanford - "Budapest Memorandum: Myths"). The UK possesses an arsenal of ~225 warheads (Trident II). **Its doctrine does not exclude first use in defense of allies, but the decision on a nuclear response to an attack on Ukraine would be political, not automatic.**

P(UK responds with nuclear weapons | Russia used) ≈ 5-15% [Model-based]. P(full nuclear war | UK responded) ≈ 40-60% [Model-based].

10.2. India and Pakistan

Conditional mechanism: Two precedents (USA + Russia) within weeks radically lower the threshold in the world's most unstable nuclear dyad. **Pakistan has a first-use doctrine against India's conventional superiority. India may reinterpret its no-first-use doctrine in light of the new norm.** Any future crisis (Kashmir, terrorism) becomes a potential trigger.

In Sagan's framing (1996/2012) [41]: Pakistan's weak institutions subject the nuclear decision to organizational and civil-military pressures, which combined with the broken taboo creates a critical scenario.

10.3. China's Role as a Potential Brake

China has significant influence over Russia and may act as a brake on the cascade. Possible conditional scenarios:

- a) **China restrains Russia** - wants to be a "responsible power" and gain moral advantage over the U.S. → P(cascade) drops by 10-20 pp
- b) **China does not restrain Russia** - weakening of the U.S. and Europe serves its strategic interests → P(cascade) unchanged

¹ France possesses an independent nuclear arsenal (~290 warheads; M51 SLBMs on Triomphant-class submarines and ASMPA-R missiles on Rafale aircraft). French doctrine is based on the concept of "dissuasion strictement suffisante" (strictly sufficient deterrence) and reserves nuclear weapon use for the defense of "intérêts vitaux de la nation" - a deliberately underspecified concept, but traditionally encompassing French territory and potentially European allies. Unlike the UK, France is not a party to the Budapest Memorandum and has no formal obligations toward Ukraine beyond the general EU/UN frameworks. However, President Macron has repeatedly signaled readiness to extend the "nuclear umbrella" over Europe (speech at the École de Guerre, 2020). The decision to use nuclear weapons rests solely with the President of the Republic - with no requirement for parliamentary or allied consultation.

c) **China exerts pressure on Pakistan** - regional stability serves Chinese interests in BRI

The Chinese factor is binary and critical.

10.4. Probability Chain

USA uses TNW on Iran → P(Russia uses in Ukraine) ≈ 30-50% → P(UK responds with nuclear weapons | Russia used) ≈ 5-15% → P(nuclear war | UK responded) ≈ 40-60%

COMBINED P(FULL NUCLEAR WAR) in this chain: 1-4% - this remains a catastrophically high number under any rational existential risk calculation.

NOTE: All P values in the cascade chain are [Model-based] and based on the author's reasoning. The chain assumes conditional independence, which is a simplification: in reality, each link changes the decision-making context of the next.

11. Potentially Inhibiting Factors

This analysis has focused on factors raising probability. For epistemic symmetry, it is necessary to identify factors potentially inhibiting the scenario's realization.

METHODOLOGICAL NOTE: Inhibiting factors have not been quantified as OR < 1.0 multipliers, because they predominantly operate in a binary manner (as necessary or exclusionary conditions), not as continuous probability modifiers. For example: resistance from U.S. military leadership either blocks the chain of command or it does not - i.e., it does not "reduce probability by 15%." Formally incorporating inhibitors into the multiplicative model would require a Bayesian network with conditional nodes (see Sec. 16.2).

11.1. Resistance of U.S. Military Leadership

U.S. generals have historically opposed nuclear weapon use (Korea, Vietnam, Persian Gulf). The chain of command requires the cooperation of the Secretary of Defense. The Pentagon's institutional culture is conservative regarding nuclear escalation. Joe Kent's resignation (intelligence, March 17, 2026) may signal broader internal resistance (AP [33]).

11.2. Israeli Security Establishment

Mossad and Shin Bet traditionally prefer surgical operations over escalation. The "Begin" Doctrine (conventional preemptive strikes) has been implemented to date without nuclear weapons. Israel's strategy relies on "nuclear ambiguity" - overt use destroys this advantage (Cohen 2010; Arms Control Association - "Wrestling With Nuclear Opacity" [18]: *"Israel will not be the first nation to introduce nuclear weapons into the region."*).

11.3. Defense Industry

Long-term defense contracts require stability and predictability. Nuclear weapon use destabilizes the global market and may trigger secondary sanctions. Some institutional and economic actors in this segment may exert a stabilizing influence if they clearly emphasize that nuclear weapon use threatens strategic predictability and long-term security interests.

11.4. Confluence of Factors as a Necessary Condition

The TNW use scenario requires the simultaneous occurrence of multiple factors: a protracted conflict, interceptor depletion, time pressure, absence of diplomatic de-escalation, and activation of domestic political factors. The absence of any single element may break the escalation chain.

METHODOLOGICAL NOTE: The analysis deliberately omits the “unpredictability” factor of the current U.S. administration. This is a variable that currently cannot be reliably quantified. It may both raise P (impulsive decisions) and lower it (unpredictability as a form of deterrence).

12. Sensitivity Analysis

Sensitivity analysis examines how changes in individual OR values affect the final probability. It allows identification of the model’s critical nodes.

12.1. Impact of Individual Variables

Test: setting each variable to OR = 1.0 (neutral) while holding others constant. Baseline: P = 3%, current P = 57%. Three variables have the greatest impact (each –17 pp when disabled): V2 Interceptor depletion, V4 China/Russia signaling, V7 Taiwan threat. **These are the model’s critical nodes.**

12.2. Multi-Variant Scenarios

- a) Pessimistic (all ORs = 2.0-2.5): P = 80%
- b) Baseline (as in the model): P = 57%
- c) Moderate (all ORs = 1.5): P = 35%
- d) Optimistic (all ORs = 1.2): P = 10%
- e) Minimal (all ORs = 1.0): P = 3%

13. Falsification Conditions

The conditions below define in advance which observations should lead to model updates. They constitute testable predictions.

13.1. Conditions Lowering P

- a) F1: Iran formally capitulates or requests negotiations (by 03/25) → –12 pp → P = 45%
- b) F2: Hezbollah announces ceasefire (by 03/28) → –5 pp → P = 52%
- c) F3: Trump declares “mission accomplished” / end of operations (by 04/05) → –15 pp → P = 42%
- d) F4: China publicly warns the U.S. against nuclear escalation → –4 pp → P = 53%
- e) F5: U.S. Congress introduces restrictions - War Powers Act (by 04/15) → –8 pp → P = 49%
- f) F6: Interceptors replenished by allies (by 04/10) → –10 pp → P = 47%
- g) F7: Operational objectives achieved - confirmed by IAEA (by 04/05) → –18 pp → P = 39%

Cumulation: If F1 + F3 + F7 occur, P drops to ~20-25%.

13.2. Conditions Raising P

- a) E1: Iran uses chemical/biological weapons → +18 pp → P = 75%
- b) E2: Iran demonstrates a nuclear device → +22 pp → P = 79%
- c) E3: Mass civilian casualties in Israel (>1,000 in a single attack) → +12 pp → P = 69%
- d) E4: Israel's interceptors fall below 20% of stockpile (by 04/05) → +10 pp → P = 67%
- e) E5: China announces exercises around Taiwan during the conflict → +6 pp → P = 63%
- f) E6: Conflict exceeds Trump's window (5 wk.) without resolution (after 04/04) → +8 pp → P = 65%

Cumulation: If E1 + E4 + E6 occur, P rises to ~85-90%.

13.3. Verification Schedule

- a) 03/25/2026: F1 (Iran capitulation) - if yes: model revision downward
- b) 03/28/2026: F2 (Hezbollah) - update
- c) **~04/05/2026: F3, F7, E6 (Trump's window) = KEY DECISION POINT**
- d) 04/10/2026: F6 (interceptors) - update
- e) 04/15/2026: F5 (Congress) - final Phase 1 verification

14. Red Team Analysis

For epistemic symmetry: a systematic list of reasons why use will NOT occur.

- a) **R1 [HIGH]:** Operation proceeding according to plan - Iran loses 90% of missile capabilities
- b) **R2 [HIGH]:** No chem/bio use by Iran - primary trigger inactive
- c) **R3 [MODERATE]:** U.S. generals historically oppose (Korea, Vietnam, Gulf)
- d) **R4 [MODERATE]:** Israeli "ambiguity" doctrine - overt use destroys the advantage
- e) **R5 [MODERATE]:** Wall Street does not want chaos - financial lobby pressure
- f) **R6 [MODERATE]:** Europe is already distancing itself - further allied losses
- g) **R7 [HIGH]:** 81 years without use - institutional and normative inertia (cf. Tannenwald 2007 - "nuclear taboo" [42])
- h) **R8 [MODERATE]:** Trump wants "victory," not "catastrophe" - success narrative requires control
- i) **R9 [LOW]:** Proliferation costs visible to the establishment (Saudi Arabia, Turkey)
- j) **R10 [MODERATE]:** U.S. public opinion opposed to wars (Reuters/Ipsos: 27-29% approval, 43% disapproval [5])

CONCLUSION: If R1 + R2 + R7 are dominant, P should be closer to 15-25% than 55-70%.

15. Revised Estimates

After incorporating the sensitivity analysis and red team analysis, the revised estimates:

- a) If the Red Team is right: P = 10-20%
- b) Optimistic scenario (OR = 1.2-1.3, brakes active): P = 20-30%
- c) Revised baseline scenario (with Red Team correction): P = 35-45%
- d) Pessimistic scenario (OR = 1.8-2.0, no brakes): P = 55-70%
- e) **Extreme scenario (E1 + E4 + E6 active): P = 75-85%**

REALISTIC RANGE: Instead of “57%,” we report 25-55%, midpoint ~40%, with full transparency regarding assumptions. The analysis’s value lies in identifying variables and mechanisms, not in specific percentages.

16. Historical Calibration and Further Research Directions

16.1. Historical Calibration

The only available reference points for model calibration are historical nuclear crises*:

- a) Cuban Missile Crisis 1962: JFK estimated the risk of war at “1/3 or even” (HKS, methodologically contested); McNamara ex post: ~30-50%
- b) Able Archer 1983: unknown, but described as “closest since Cuba”
- c) Iran 2026 (this model): 25-55%, midpoint ~40%

The comparison suggests that the current crisis falls within a range comparable to the Cuban Missile Crisis. However, it should be emphasized that even historical estimates are retrospective and methodologically contested - nobody had then (or has now) a reliable predictive model.

***The author deliberately did not consider the use of nuclear weapons by the U.S. during World War II, as this precedent could reduce this analysis from estimating probability - i.e., the question “whether?” - to posing a different question, namely: “when?”**

16.2. Further Research Directions* (beyond the scope of this analysis)

The author is aware of the methodological limitations of this analysis. Deepening the topic would require actions exceeding the competencies and resources of a single analyst:

- a) **Formal expert elicitation:** A panel of experts (intelligence analysts, nuclear doctrine specialists, Cold War historians) using the Delphi method or - preferably - Cooke’s classical model (Structured Expert Judgment / SEJ), which uses calibration questions (seed questions) to weight experts by accuracy. See: Cooke (1991), “Experts in Uncertainty” [19]; Cooke & Goossens (2008) [20].
- b) **Bayesian network with correlations:** Transition from the multiplicative OR model to a full Bayesian network (tools: GeNIe, Netica, PyMC) with explicit modeling of inter-variable dependencies. Proposed minimal structure: Conflict >4–6 wk. → Reputational pressure; Interceptor status → Defensive breakthrough → Existential threat; CBRN → Trigger; China/Russia signaling → Pressure; Brakes → P(TNW use). The network allows (a) explicit dependency modeling, (b) separation of “hard” from “operational” nodes, (c) updating as data flows in.
- c) **Validation of the baseline on historical data:** Analysis of all nuclear crises 1945-2025 using the International Crisis Behavior Project (ICB, Duke University / University of Southern California) database [21], identifying predictive variables and calibrating the baseline P_0 . Calibration should employ the Brier score.
- d) **Analysis of institutional mechanisms:** Study of decision-making processes (U.S. chain of command, civil-military relations, role of the Secretary of Defense) with reference to the literature (Sagan, Feaver).

*** Due to the potentially approaching window for TNW use, the research directions listed in this section are more theoretical than practically applicable. Firstly, because if the TNW use scenario does not materialize in the near term, there is little chance it will come to fruition - at least as far as use by the U.S. is concerned, since the "existential threat" to Israel may retain its triggering power for a considerably longer period.**

16.3. Suggested Verification Questions

The questions below serve to strengthen the document without changing its purpose. Each indicates an area where the current version requires further clarification:

- a) **VQ1:** How does the author define “tactical nuclear weapon use”? → Answer: see Section 2.
- b) **VQ2:** Where does the 3% prior at week 6 come from and how is it validated? → Answer: see Section 6, as well as 8.4.
- c) **VQ3:** Which nodes are interdependent and how strongly (e.g., “Taiwan” → “signaling” → “reputation”)? → Partial answer: methodological note in Sec. 5.2 and Sec. 8.1; a full answer requires a Bayesian network (Sec. 16.2).
- d) **VQ4:** What are the criteria for distinguishing “hard data” from “operational claims”? → Answer: source classification [Type] introduced in Section 2.
- e) **VQ5:** How to operationally update “IAEA will confirm” within a timeframe of weeks, given that the IAEA itself indicates knowledge limitations without inspection? → Answer: open problem. The IAEA (Grossi, 03/18/2026 [2]) explicitly states lack of knowledge about Isfahan’s status without inspection. Condition F7 may be impossible to fulfill in a short timeframe.

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